■# PAPER<i>354 — D</i>Universe 5th Factor: Spatial Curvature Completion of the 4-Factor Chain (PAPER_296)

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Session: *96*

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Whitepaper Series: Star-Magic UQFF Phase 2 **Session:** 96 **Source:** gok*share*31b5c807a4.txt (Supplemental Gap Analysis Block) **Classification:** FIRST UQFF 5th factor spatial curvature term for D*universe; completes PAPER*296 chain **Author:** Daniel T. Murphy

<!-- UQFF constants: κ = 5.0e-4 day^{-1}, [SSq] = 0.57, M_UQFF = 1.43e1 TeV -->

Abstract

PAPER*296 established a 4-factor chain for the UQFF Universe expansion parameter D*universe. This paper adds the mandatory 5th factor: a spatial curvature correction (1 + k·rc^2*), where k is the curvature constant and r*c is the Friedmann comoving curvature radius. The complete 5-factor D*universe is now: D*universe = [4 prior factors] × (1 + k·rc^2*). For a flat universe (k = 0), the 5th factor = 1 and PAPER*296 is recovered. For non-flat models, this term accounts for the deviation of cosmic spatial geometry from the Minkowski approximation used in earlier UQFF distance calculations.

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2. Core Physics

2.1 Complete 5-Factor D_universe

```
D_{\rm universe} = D_1 \cdot D_2 \cdot D_3 \cdot D_4 \cdot (1 + k_{\rm curv} \cdot r_c^2)
```

where factors D*1 through D*4 were established in PAPER_296 (Session 91) and the new 5th factor is:

```
D_5 = 1 + k_{\rm curv} \cdot r_c^2
```

2.2 Curvature Parameter k

The Friedmann equation curvature:

```
k_{\rm curv} = \frac{(H_0^2 / c^2)(\Omega_{\rm total} - 1)}{1}
```

For the Planck 2018 constraint Ω_total = 1.0007 ± 0.0019:

```
k_{\rm curv} \approx 0.0007 \cdot \frac{H_0^2}{c^2} \approx 5.3 \times 10^{-54}\
\mathrm{m}^{-2}
```

2.3 Curvature Correction at Cosmological Scale

At r*c = Hubble radius (R*H = c/H_0 ≈ 1.37×10^{2}■ m):

```
D_5 = 1 + 5.3\times 10^{-54} \times (1.37\times 10^{26})^2 = 1 + 5.3\times 10^{-54} \times
1.88\times 10^{52}
```

```
D_5 = 1 + 0.001 = 1.001
```

A 0.1% correction — detectable by next-generation CMB experiments (e.g., CMB-S4, LiteBIRD).

2.4 Near-Flat Expansion Series

For small curvature ($k_{curv} \cdot r_c^2$ « 1):

```
D_5 \approx 1 + k_{\rm curv} r_c^2 - \frac{(k_{\rm curv} r_c^2)^2}{2} + \ldots
```

The leading correction is linear in both k and r_c^2.

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3. Key Values

Quantity	Formula	Value
k_curv	Planck 2018 constraint	~5.3×10^-■■ m^-2
r_c (Hubble radius)	c/H_0	1.37×10^2■ m
D_5 (Hubble scale)	$1 + k \cdot r_c^2$	1.001
D_5 (flat limit)	k = 0	1.000
PAPER_296 factors	$D_1 \times D_2 \times D_3 \times D_4$	Previously computed

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4. Physical Significance

The 5th factor completes the UQFF $D_{universe}$ chain, which now accounts for: (1) vacuum buoyancy scale factor, (2) string rotation expansion term, (3) Hubble flow scale, (4) charge-reactivity expansion coupling, and (5) spatial curvature geometry. The chain $D_{universe} = D_1 \times D_2 \times D_3 \times D_4 \times D_5$ represents the most complete UQFF treatment of cosmic expansion parameters. The 0.1% curvature correction at Hubble scale sets the signal size for CMB observational tests: future CMB-S4 measurements of the spatial curvature power spectrum should detect D_5 deviations from unity at the ~0.05% level.

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5. Deduplication Note

vs. PAPER_296: PAPER_296 derived the 4-factor chain; PAPER_354 adds the mandatory spatial curvature 5th factor.

Unique: The $(1 + k \cdot r_c^2)$ form is new — no earlier UQFF paper included spatial curvature directly in $D_{universe}$.

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6. Classification

Physics Territory: FIRST UQFF $D_{universe}$ spatial curvature 5th factor; completes PAPER_296 four-factor chain **Scale:** Cosmological (Hubble radius; universal) **CP Implementation:** `DUniverseSpatialCurvatureFifthFactorCalculator` (CondensedPhysics3.py, Session 96)
